LU Talk (Dec, 2025)

Language Understanding λ(μ) Laboratory

Beyond Dense Vectors: Unleashing the Power of Hyperdimensional Computing

FOR DECADES, THE VON NEUMANN ARCHITECTURE HAS DEFINED COMPUTING THROUGH BINARY LOGIC AND PRECISE INSTRUCTION SETS. BUT AS WE PUSH THE BOUNDARIES OF ARTIFICIAL INTELLIGENCE, A FUNDAMENTAL QUESTION ARISES: CAN WE BUILD MACHINES THAT REASON MORE LIKE THE HUMAN BRAIN—ROBUST, HOLOGRAPHIC, AND INHERENTLY NOISE-TOLERANT?

THIS TALK EXPLORES HYPERDIMENSIONAL COMPUTING (HDC), A PARADIGM SHIFT THAT REPLACES LOW-DIMENSIONAL DENSE VECTORS WITH HYPERDIMENSIONAL VECTORS (HVS)—MASSIVE, RANDOM VECTORS TYPICALLY SPANNING 10,000 DIMENSIONS. UNLIKE TRADITIONAL ARCHITECTURES, HDC OPERATES ON A VECTOR-SYMBOLIC ARCHITECTURE (VSA), WHERE SYMBOLS ARE REPRESENTED AS UNIQUE POINTS IN A VAST HYPERSPACE. THROUGH AN ELEGANT ALGEBRA OF BINDING, BUNDLING, AND PERMUTATION, HDC ALLOWS US TO ENCODE COMPLEX DATA STRUCTURES—FROM HASH TABLES AND SEQUENCES TO KNOWLEDGE GRAPHS AND SYNTACTIC TREES—INTO A SINGLE, FIXED-DIMENSION VECTOR.

WE WILL DIVE INTO THE CORE MECHANICS OF HDC, INCLUDING:

- THE POWER OF VSAS: A COMPARISON OF POPULAR MODELS LIKE MAP, HRR, FHRR, AND BSC, AND HOW THEY BALANCE COMPLEXITY WITH INFORMATION CAPACITY.
- NEURO-SYMBOLIC INTEGRATION: HOW HDC BRIDGES THE GAP BETWEEN SYMBOLIC LOGIC AND DEEP LEARNING, ENHANCING TRANSFORMERS AND ENABLING SINGLE-PASS MACHINE LEARNING.
- REAL-WORLD APPLICATIONS: FROM ULTRA-FAST UNSUPERVISED TOPIC SEGMENTATION (HYPERSEG) TO ROBUST LANGUAGE RECOGNITION AND GRAPH-BASED MEMORIZATION (GRAPHD).

Dr. Prachya Boonkwan

PRACHYA BOONKWAN IS A LECTURER AT SIRINDHORN INTERNATIONAL INSTITUTE OF TECHNOLOGY (SIIT), THAMMASAT UNIVERSITY, THAILAND, SINCE 2025. FORMERLY, HE WAS A SENIOR RESEARCHER AND THE DIRECTOR OF LANGUAGE AND SEMANTIC TECHNOLOGY RESEARCH TEAM, ARTIFICIAL INTELLIGENCE RESEARCH UNIT, NATIONAL ELECTRONICS AND COMPUTER TECHNOLOGY CENTER, SINCE 2005. HIS TOPICS OF INTEREST INCLUDE: GRAMMAR INDUCTION, STATISTICAL PARSING, NATURAL LANGUAGE PROCESSING, FORMAL SYNTAX, AND QUANTUM COMPUTING. HE WAS ALSO INVITED TO GIVE LECTURES IN 12 UNIVERSITIES IN SOUTHEAST ASIA, KEYNOTE SPEECHES ON NATURAL LANGUAGE PROCESSING AND DEEP LEARNING IN SEVERAL ACADEMIC CONFERENCES, AND A TEDX-LIKE TALK ON THE AFOREMENTIONED TOPICS.

EDUCATION BACKGROUND:

- PHD IN INFORMATICS (COMPUTATIONAL LINGUISTICS): THE UNIVERSITY OF EDINBURGH, UNITED KINGDOM
- MASTERS OF ENGINEERING (COMPUTER ENGINEERING): KASETSART UNIVERSITY, THAILAND
- BACHELOR OF ENGINEERING (COMPUTER ENGINEERING): KASETSART UNIVERSITY, THAILAND

DATE: 20 Dec 2025 (SAT)

Time: 11:00 to 12:00 (Thailand Time)

Meeting Link:https://meeting-nstda.webex.com/wbxmjs/joinservice/sites/meeting-nstda/meeting/download/26665c25f99349fa8dd7e310f211c63f?protocolUID=687e2f72a665f3c218f5b89375a93b2b